

MT-1003 Calculus and Analytical Geometry

Assignment # 1

Note: Last date for the submission of assignment is 08-09-2022.

Q 1. The product of two integers is at most 28 and their sum is 16. What are the numbers?

Q 2. The difference of the squares of two consecutive even integers is less than 72. What are the numbers?

Q 3. Sara needs to make a garden plot which has an area less than 22 ft^2 . The length should be 3 ft longer than the width. What are the possible dimensions of the plot?

Q 4. An object is launched from ground level directly upward at a rate of 144 feet/sec. The equation for object's height is $y = -16x^2 + 144x$. What values of x is the object at or above a height of 288 feet?

Q 5. Ali wants to build a rectangular garden with a fence. He wants the length of the garden is 7 feet larger than the width. If area of the garden must be less than 20 square meters. What are the possible dimensions of his garden?

Q 6. The cost function $C(x)$ and revenue function $R(x)$ for producing x units of certain product are given by

$$C(x) = 5x + 350, \quad R(x) = 50x - x^2$$

Find the value of x that yield a profit.

Q 7. Keith has \$500 in a savings account at the beginning of the summer. He wants to have at least \$200 in the account by the end of the summer. He withdraws \$25 each week for food, clothes, and movie tickets.

- Write an inequality that represents Keith's situation.
- How many weeks can Keith withdraw money from his account? Justify your answer.

Q8. Find the solution of given inequalities.

i. $|x| + |x - 1| > 1$

ii. $|2x + 5| > |2 - 5x|$

iii. $\frac{x-1}{2} - \frac{1}{x} > \frac{4}{x} + 5$

iv. $x^{-2} - 4x^{-1} - 5 \geq 0$